STASIS · GAP ANALYSIS · 23 OCTOBER 2026

What Has Not Been Done

Measured against *STASIS Build Guide: A Phase-by-Phase Guide*, checked against the repository and the Control Room server.

> **The building is finished. The measuring has barely begun.**
>
> Every component exists and works: the controller, the BPF scheduler, the dashboard, the test harness. One full end-to-end run proves the idea is sound — 7 heavy jobs finished all 420 passes with nothing killed, where plain Linux managed 4 passes and gave up.
>
> What is missing is the evidence that STASIS is *better* than the alternatives. That evidence is the result. None of it can be gathered on the Windows machine — it all needs the Hyper-V Ubuntu VM with `sched_ext`.

1 The evaluation campaign

This is the largest gap. `eval/campaign.sh` expects six configurations. Three have never been run even once — and they are precisely the competitors the project claims to beat.

CONFIGURATION	WHAT IT IS	RUNS REQUIRED	RUNS DONE	STATUS
`eevdf`	Stock Linux, the baseline	40	2	Barely started
`stasis`	Full STASIS	40	1	Barely started
`ctrl`	Controller only	15	1	Barely started
`oomd`	Linux's own memory killer	15	0	Never run
`simple`	Stock `scx_simple` scheduler	15	0	Never run
`thrashprotect`	The closest existing tool	15	0	Never run

Beyond the missing configurations, the runs that do exist are thin: **one run per cell** where the pre-registration demands five, and only at **N = 2 and N = 7** where the plan covers N = 1 to 8.

2 Not one hypothesis is scored

The server's own score table returns `"result":"MISSING"` for every row — H1, H1c, H2, H2c, H3, H3c, H4 and H5. That is why the Results view reads *No data* all the way down.

These are not failures. They have not been attempted. The decision rule is strict: a hypothesis passes only if every STASIS run beats every baseline run. With no baseline runs, nothing can pass or fail.

3 Gates not yet passed

GATE	WHAT IT REQUIRES	WHERE IT STANDS
Gate 1	The EEVDF throughput curve over N = 1…8, five runs each, plus Figure 1	Two runs total (N = 2, N = 7). No figure exists.
Gate 2	N = 8 under EEVDF + stasisd, all jobs finishing, zero kills	Achieved at N = 7, not N = 8.
Gate 3	The N = 1…8 sweep repeated under `scx_stasis` with the controller off	No such runs exist. The only labels recorded are `eevdf`, `ctrl` and `stasis`.
Gate 4	Three consecutive N = 8 runs of full STASIS, decision log showing both `psi` and `pff` events	One run, at N = 7. No `pff` event has ever been recorded first.

4 Two frozen values are still blank

In `docs/00-prereg.md`, two parameters that were supposed to be locked before any Phase 5 run remain placeholders:

- Line 15 — `PASSES` is still `__`
- Line 27 — `PFF_HI` is still `____`

`PFF_HI` is the page-fault-frequency threshold the entire early-warning mechanism depends on. H5 cannot be answered while it is empty, and it cannot be filled without the Phase 1 calibration runs on the VM.

5 The early warning has never won a race

The central novelty of STASIS is that the scheduler should notice trouble *before* the kernel's own pressure signal does — faults per millisecond of a job's own CPU time, which PSI cannot measure.

In both recorded controlled runs, `first_src` is `psi`. The kernel noticed first, every time. **The mechanism that makes this project novel has not yet demonstrated its advantage once.**

6 Four fixes never run on a kernel

Reading the rehearsal logs exposed four further problems, all since corrected in code:

- Sets that already fit were halved while reloading (deviation 10)
- The probe waited behind a batch job's slice — p99 of 16.0 ms against 8.1 ms controller-only (deviation 11)
- Runs inherited the previous run's PSI reading (deviation 12)
- Fractional byte counts were sent to `memory.reclaim`, which the kernel rejects

These fixes are covered by `tests/test_controller.sh` and the simulator, and `scx_stasis` still compiles. **But none of them has run on an actual kernel.** The next Phase 2–4 runs will be their first real test.

7 Sensitivity analysis — not started

The plan calls for showing that the conclusion does not hinge on one lucky setting, by sweeping at N = 8:

- Rotation quantum: 5 s, 15 s, 30 s
- Trigger threshold: 50 ms, 100 ms, 200 ms per 1 s window

No such runs exist.

8 Write-up and demo

- **Figures F1 to F5** — none exist; there is no figures directory anywhere in the repository.
- **The backup demo video** — not recorded. The plan asks for one in case the live VM misbehaves during the defence.

`docs/demo.md`, the demo script and rehearsal questions, is written.

9 One number worth investigating

On the full-STASIS run, Jain's fairness index is **0.143**. For seven jobs that is exactly 1/7 — the mathematical floor, meaning one job made all the progress in the measurement window and the other six made none. Controller-only scored 0.428 on the same N.

Fairness is H3, the claim that distinguishes STASIS from thrash-protect. The window is narrow (first pressure event to first job finishing) so this may be an artefact, but it should be understood before the campaign rather than after.

10 For contrast — what is done

- All code: the controller, the BPF scheduler and its loader, the sampler, the run harness, the analysis tools, the regression tests.
- The Control Room dashboard — four working views, and not asked for by the guide at all.
- Phase 0 provisioning, verified end to end on Ubuntu 26.04 / kernel 7.0 in 3 min 41 s.
- One passing rehearsal of each of Phases 1 through 4, including the fault-count correctness check (17,645 = 17,645).
- The pre-registration document, the deviation record, and the demo script.

11 Evidence

Every statement above comes from the repository or from the running Control Room server, not from the README's claims.

CLAIM	SOURCE
Four runs, three configurations	`/api/results` and `docs/rehearsal/results/metrics.csv`
Six configurations expected	`eval/campaign.sh`, line 32
All hypotheses MISSING	`/api/results`, score array
Blank `PASSES` and `PFF_HI`	`docs/00-prereg.md`, lines 15 and 27
`first_src` is `psi` in both runs	`metrics.csv`, column `first_src`
Jain 0.143 and 0.428	`metrics.csv`, column `jain`
No figures directory	Filesystem search of the repository

The four recorded runs, in full

CONFIG	N	STATUS	THROUGHPUT	PASSES	KILLS	JAIN	PROBE P99	FIRST PARK
eevdf	2	OK	6.961/s	120	0	—	5.5 ms	—
eevdf	7	**TIMEOUT**	0.013/s	4	0	0.286	11.0 ms	—
ctrl	7	OK	0.695/s	420	0	0.428	8.1 ms	2.5 s
stasis	7	OK	1.496/s	420	0	0.143	16.0 ms	24.7 s

A caution on reading these numbers: they come from a rehearsal VM emulated in software, with swap running at roughly 3 MB/s. They demonstrate that the mechanism works; they are not results. The probe p99 of 16.0 ms in particular is the pre-fix figure — deviation 11, since corrected but not yet re-measured.